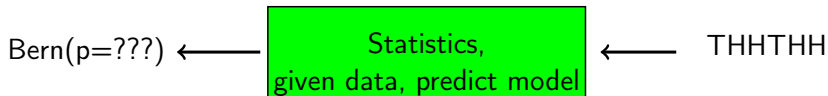
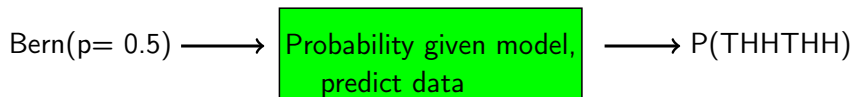


Maximum-Likelihood Estimation

Probability vs Statistics



Likelihood Estimation

I give you and your friend each 6 minutes with a coin with unknown probability of heads p . whoever has the closest estimate will win. what do you do in your 6 minutes and what do you give as your estimate?

We will toss the coin many times, record # of Heads and # of tails, then our estimate for

$$P(\text{Head}) = \frac{\# \text{ heads}}{\# \text{ heads} + \# \text{ of tails}}$$



Let's say you saw 6 heads and 2 tail, you will tell $\hat{p} = \frac{3}{4}$. How can you objectively argue that this is the best estimate?

Is there some objective function that it maximizes?



$$P(\text{having the above pattern}) = p^6 (1-p)^2$$

This a function of p and we are looking for the value of p that maximizes this function. because we want the highest chance that this pattern be observed.

Likelihood

You assume a model (Bernoulli in our case) with unknown parameter θ , and receive iid samples $x = (x_1, x_2, \dots, x_n) \sim \text{Bern}(\theta)$. The likelihood of the data given a parameter θ is

$$\begin{aligned} L(X, \theta) &= P(\text{seeing data}; \theta) \\ &= P(x_1, \dots, x_n; \theta) = \prod_{i=1}^n p_X(x_i; \theta) \end{aligned}$$

we can multiply due to independence

Likelihood

Definition (Realization/Sample:)

A realization/sample x of a random variable X is the value that is actually observed.

Definition (Likelihood Function)

Let $x = (x_1, x_2, \dots, x_n)$ be iid realization from probability mass function $p_X(t; \theta)$ if X is discrete, or from a density $f_X(t; \theta)$ if X is continuous, where θ is a parameter or vector of parameters. We define the likelihood of x as the probability of seeing the data.

$$\text{If } X \text{ is discrete} \quad L(x, \theta) = \prod_{i=1}^n p_X(x_i; \theta)$$

$$\text{If } X \text{ is continuous} \quad L(x, \theta) = \prod_{i=1}^n f_X(x_i; \theta)$$

Maximum likelihood estimation (Bernoulli)

$$L(\text{HHHTHHHTH}; \theta) = \theta^6(1 - \theta)^2 = \theta^6[1 + \theta^2 - 2\theta] = \theta^6 + \theta^8 - 2\theta^7$$

$$\Rightarrow L' = 6\theta^5 + 8\theta^7 - 14\theta^6 = 2\theta^5[3 + 4\theta^2 - 7\theta]$$

$$L' = 2\theta^5(\theta - 1)(4\theta - 3) = 0 \Rightarrow \theta = 0 \text{ or } \theta = 1 \text{ or } \theta = \frac{3}{4}$$

$$L'' = 30\theta^4 + 56\theta^6 - 84\theta^5 = 2\theta^4[15 + 28\theta^2 - 42\theta]$$

$$L''(\theta = 0) = 0 \Rightarrow \text{second derivative test is inconclusive. [see next page for detail]}$$

$$L''(\theta = 1) = 2[15 + 28 - 42] = 2 > 0$$

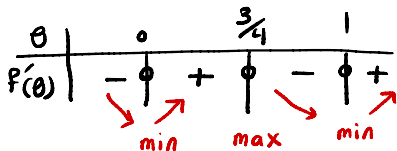
$$L''(\theta = \frac{3}{4}) = 2\left(\frac{3}{4}\right)^4 \left[15 + 28 \cdot \frac{9}{16} - 42 \cdot \frac{3}{4}\right] = 2\left(\frac{3}{4}\right)^4 \left[-\frac{3}{4}\right] = -2\left(\frac{3}{4}\right)^5 < 0$$

$\theta = 0, 1, \frac{3}{4}$ are critical values but only $\hat{\theta} = \frac{3}{4}$ makes the likelihood

function maximum, since $L''(\theta = \frac{3}{4}) < 0$. Hence $\hat{\theta} = \frac{3}{4}$ is our estimate.

To clarify which type of point $\theta=0$ is for function $f(\theta)=\theta^6(1-\theta)^2$, you can use the sign of the first derivative.

$$\begin{aligned} f'(\theta) &= 2\theta^5[3+4\theta^2-7\theta] \\ &= 2\theta^5(\theta-1)(4\theta-3) \end{aligned}$$



Maximum likelihood estimation (Bernoulli)

$$L(HHHTHHHTH|\theta) = \theta^6(1 - \theta)^2$$

Maximum likelihood Estimation(MLE)

Let $x = (x_1, x_2, \dots, x_n)$ be iid realization from probability mass function $p_x(t; \theta)$ if X is discrete, or from a density $f_X(t; \theta)$ if X is continuous, where θ is a parameter or vector of parameters. We define the maximum likelihood estimator $\hat{\theta}_{MLE}$ of θ to be the parameter which maximizes the likelihood (or equivalently, the log-likelihood).

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta} L(x; \theta) = \operatorname{argmax}_{\theta} \ln L(x; \theta)$$

Steps to find MLE

- 1- Compute the likelihood and log-likelihood of data.
- 2- Take derivative with respect to θ and set the derivative to 0. Solve the equation.
- 3- Verify $\hat{\theta}_{MLE}$ is indeed a local maximizer by checking the second derivative at $\hat{\theta}_{MLE}$ is negative (if θ is a single matrix), or the Hessian is negative semi-definite (if θ is a vector of parameters)

Example of MLE: Poisson

Eg1. Let's say $x_1, x_2, \dots, x_n$ are iid samples from $Poi(\theta)$. (might look like $x_1 = 3, x_2 = 5, x_3 = 4$, etc). What's MLE of θ ?

$$X \sim Poi(\theta) \Rightarrow P(X=k) = \bar{e}^{-\theta} \frac{\theta^k}{k!}$$

$$L(X; \theta) = L(x_1, x_2, \dots, x_n; \theta) = \prod_{i=1}^n \bar{e}^{-\theta} \frac{\theta^{x_i}}{x_i!}$$

Recall $\ln(ab) = \ln a + \ln b$

$\ln\left(\frac{a}{b}\right) = \ln a - \ln b \rightarrow$

$$\ln L(X; \theta) = \sum_{i=1}^n \ln\left(\bar{e}^{-\theta} \frac{\theta^{x_i}}{x_i!}\right)$$

$$\ln L(X; \theta) = \sum_{i=1}^n (\ln \bar{e}^{-\theta} + \ln \theta^{x_i} - \ln(x_i!))$$

$$\ln L(X; \theta) = \sum_{i=1}^n (-\theta + x_i \ln \theta - \ln(x_i!))$$

$$\frac{\partial}{\partial \theta} \ln L(X; \theta) = \sum_{i=1}^n \left(-1 + \frac{x_i}{\theta} \right) = -n + \frac{\sum_{i=1}^n x_i}{\theta}$$

$$-n + \frac{\sum_{i=1}^n x_i}{\hat{\theta}} = 0 \Rightarrow \hat{\theta} = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}$$

$$\frac{\partial^2}{\partial \theta^2} \ln L(X; \theta) = - \frac{\sum_{i=1}^n x_i}{\theta^2} \quad \frac{\partial^2}{\partial \theta^2} \ln L(X; \hat{\theta}) = - \frac{n^2}{\sum_{i=1}^n x_i} < 0$$

Hence $\ln(L(x; \theta))$ is maximum at $\hat{\theta} = \bar{x}$

Meaning a good estimate for θ in the case of poisson is sample mean.

Eg2. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from the distribution with probability density function

$$f_X(x) = f_X(x; \theta) = (\theta - 1)^2 \frac{\ln x}{x^\theta} \quad \theta > 1, x > 1$$

$$L(x; \theta) = \prod_{i=1}^n f_X(x_i; \theta) = \prod_{i=1}^n (\theta - 1)^2 \frac{\ln x_i}{x_i^\theta}$$

$$\Rightarrow \ln(L(x; \theta)) = \sum_{i=1}^n \left[2 \ln(\theta - 1) + \ln(\ln x_i) - \theta \ln x_i \right]$$

$$\ln(L(x; \theta)) = 2n(\ln(\theta - 1)) + \sum_{i=1}^n \ln(\ln x_i) - \theta \sum_{i=1}^n \ln x_i$$

$$\frac{\partial}{\partial \theta} (\ln(L(x; \theta))) = 0$$

$$\frac{\partial}{\partial \theta} (\ln(L(x; \theta))) = \frac{2n}{\theta - 1} - \sum_{i=1}^n \ln x_i$$

$$\frac{1}{\hat{\theta} - 1} = \frac{\sum_{i=1}^n (\ln x_i)}{2n} \Rightarrow \hat{\theta} - 1 = \frac{2n}{\sum_{i=1}^n (\ln x_i)}$$

$$\hat{\theta} = 1 + \frac{2n}{\sum_{i=1}^n (\ln x_i)}$$

We check the second derivative to make sure $\hat{\theta}$ is a maximizer for $\ln(L(x; \theta))$ function.

$$\frac{\partial^2}{\partial \theta^2} (\ln(L(x; \theta))) = \frac{-2n}{(\theta - 1)^2} < 0. \text{ This is always negative.}$$

$\hat{\theta} = 1 + \frac{2n}{\sum_{i=1}^n (\ln x_i)}$ is the maximum likelihood estimation,

meaning that it's the parameter which with that we will have the highest probability.

Example of MLE: Normal

Eg3 is only for your information, it will not be used for quizzes.

Eg3. Let $X_1, X_2, \dots, X_n$ be a random sample of size n from $N(\theta_1, \theta_2)$, where $\Omega = \{(\theta_1, \theta_2) : -\infty < \theta_1 < \infty, -\infty < \theta_2 < \infty\}$. That is, here we let $\theta_1 = \mu$ and $\theta_2 = \sigma^2$.

Obtain the maximum likelihood estimator of θ_1 , i.e. $\hat{\theta}_1$, and of θ_2 , i.e. $\hat{\theta}_2$

$$f_x(x) = \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x-\theta_1)^2}{2\theta_2}}$$

$$L(x_1, x_2, \dots, x_n; \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\theta_2}} e^{-\frac{(x_i-\theta_1)^2}{2\theta_2}}$$

$$\ln(L(x_1, x_2, \dots, x_n; \theta)) = \sum_{i=1}^n \ln\left(\frac{e^{-\frac{(x_i-\theta_1)^2}{2\theta_2}}}{\sqrt{2\pi\theta_2}}\right)$$

$$\ln(L(x_1, x_2, \dots, x_n; \theta)) = \sum_{i=1}^n \ln \left(\frac{e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}}}{\sqrt{2\pi\theta_2}} \right)$$

$$= \sum_{i=1}^n \left[\ln \left(e^{-\frac{(x_i - \theta_1)^2}{2\theta_2}} \right) - \ln \sqrt{2\pi\theta_2} \right]$$

$$\ln(L(x_1, x_2, \dots, x_n; \theta)) = - \sum_{i=1}^n \frac{(x_i - \theta_1)^2}{2\theta_2} - \frac{n}{2} \ln(2\pi\theta_2)$$

$$\frac{\partial}{\partial \theta_1} \ln(L(x_1, x_2, \dots, x_n; \theta)) = \sum_{i=1}^n \frac{(x_i - \theta_1)}{\theta_2} = \frac{1}{\theta_2} \left(\sum_{i=1}^n x_i - n\theta_1 \right)$$

$$\frac{\partial}{\partial \theta_1} \ln(L(x_1, x_2, \dots, x_n; \hat{\theta})) = 0 \Rightarrow \hat{\theta}_1 = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}_n$$

$$\frac{\partial}{\partial \theta_2} \ln(L(x_1, x_2, \dots, x_n; \theta)) = \left(\sum_{i=1}^n \frac{(x_i - \theta_1)^2}{2} \right) \frac{1}{\theta_2^2} - \frac{n}{2\theta_2}$$

$$\left(\sum_{i=1}^n \frac{(x_i - \hat{\theta}_1)^2}{2} \right) \frac{1}{\hat{\theta}_2^2} - \frac{n}{2\hat{\theta}_2} = 0$$

$$\Rightarrow \left(\sum_{i=1}^n (x_i - \hat{\theta}_1)^2 \right) \frac{1}{\hat{\theta}_2^2} = n$$

$$\Rightarrow \hat{\theta}_2 = \frac{\sum_{i=1}^n (x_i - \hat{\theta}_1)^2}{n} = \frac{\sum_{i=1}^n (x_i - \bar{x}_n)^2}{n}$$